



**Introduction to  
Internet of Things  
Assignment-Week 5**

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions: 15**

**Total marks: 15 X 1= 15**

**QUESTION 1:**

**What does interoperability allow IoT devices to do?**

- a. Communicate and exchange data
- b. Work only offline
- c. Use only one protocol
- d. Disable other devices

**Correct Answer: a. Communicate and exchange data**

**Detailed Solution:** Interoperability enables different IoT devices and systems to communicate and exchange data.

See Lecture 21 (Interoperability in Internet of Things) @ 03:30

**QUESTION 2:**

**Syntactic interoperability deals with:**

- a. Message format
- b. Battery size
- c. Device colour
- d. Sensor weight

**Correct Answer: a. Message format**

**Detailed Solution:** Syntactic interoperability concerns the format of messages exchanged between devices and users.

See Lecture 21 (Interoperability in Internet of Things) @ 17:01



**QUESTION 3:**

**What does UMB stand for?**

- a. Universal Middleware Bridge
- b. Unified Mobile Board
- c. Universal Memory Block
- d. User Management Bridge

**Correct Answer: a. Universal Middleware Bridge**

**Detailed Solution:** UMB stands for Universal Middleware Bridge. It helps different middleware-based devices work together.

See Lecture 21 (Interoperability in Internet of Things) @ 25:42

**QUESTION 4:**

**What is the operating voltage of the Arduino UNO?**

- a. 3.3 V
- b. 5 V
- c. 9 V
- d. 12 V

**Correct Answer: b. 5 V**

**Detailed Solution:** The Arduino UNO operates at 5 volts.

See Lecture 22: Introduction to Arduino Programming – Part I @ 06:57.



**QUESTION 5:**

**What is a program written in the Arduino IDE called?**

- a. Packet
- b. Sketch
- c. Frame
- d. Query

**Correct Answer: b. Sketch**

**Detailed Solution:** A program created and uploaded using the Arduino IDE is called a sketch.

See Lecture 22 (Introduction to Arduino Programming - Part I) @ 12:12

**QUESTION 6:**

**Which interface is mainly used to connect an Arduino UNO to a computer?**

- a. HDMI
- b. USB
- c. VGA
- d. Ethernet

**Correct Answer: b. USB**

**Detailed Solution:** The Arduino UNO uses a USB interface to connect to a computer. The USB connection can be used to upload programs to the board and also provide power during development.

See Lecture 22: Introduction to Arduino Programming – Part I @ 07:26



**QUESTION 7:**

**What is the conventional starting index of an array in Arduino programming?**

- a. -1
- b. 0
- c. 1
- d. 2

**Correct Answer: b. 0**

**Detailed Solution:** As in C and C++, Arduino arrays use zero-based indexing. Therefore, the first element is stored at index 0.

See Lecture 23 (Introduction to Arduino Programming - Part II) @ 04:08

**QUESTION 8:**

**Which Arduino function is used to generate a pseudo-random number within a specified range?**

- a. random()
- b. pinMode()
- c. digitalWrite()
- d. attachInterrupt()

**Correct Answer: a. random()**

**Detailed Solution:** The random() function generates a pseudo-random number in a specified range. For example, random(max) produces a value within the range defined by max.

See Lecture 23 (Introduction to Arduino Programming - Part II) @ 09:19



**QUESTION 9:**

**In the traffic-control example, which LED is switched ON first to represent the GO signal?**

- a. Red LED
- b. Yellow LED
- c. Green LED
- d. All three LEDs simultaneously

**Correct Answer: c. Green LED**

**Detailed Solution:** The traffic() function first sets the green LED HIGH and displays the GO message. After a delay, it switches to yellow and then red.

See Lecture 23 (Introduction to Arduino Programming - Part II) @ 14:34

**QUESTION 10:**

**Which quantities are measured by a DHT sensor?**

- a. Temperature and humidity
- b. Distance and speed
- c. Light and sound
- d. Pressure and altitude

**Correct Answer: a. Temperature and humidity**

**Detailed Solution:** A DHT sensor measures the temperature and humidity of its surroundings.

See Lecture 24: Integration of Sensors and Actuators with Arduino – Part I @ 04:37.



**QUESTION 11:**

What does a sensor convert a physical quantity into?

- a. A software program
- b. Mechanical motion
- c. A network address
- d. An electrical signal

**Correct Answer:** d. An electrical signal

**Detailed Solution:** A sensor is an electronic element that converts a physical quantity or measurement into an electrical signal.

See Lecture 24: Integration of Sensors and Actuators with Arduino – Part I, Sensors section.

**QUESTION 12:**

In the DHT sensor sketch, `delay(2000)` causes successive readings to be printed after approximately how much time?

- a. 0.2 seconds
- b. 1 second
- c. 2 seconds
- d. 20 seconds

**Correct Answer:** c. 2 seconds

**Detailed Solution:** The `delay()` function takes time in milliseconds. Therefore, `delay(2000)` introduces a delay of 2000 milliseconds, which equals 2 seconds.

See Lecture 24 (Integration of Sensors and Actuators with Arduino - Part I) @ 13:34



**QUESTION 13:**

State whether the following statement is True or False.

**Statement:** A servo motor is a low-precision motor used only for linear motion.

- a. True
- b. False

**Correct Answer:** b. False

**Detailed Solution:** A servo motor is a high-precision motor that provides controlled rotary motion.

**See Lecture 25: Integration of Sensors and Actuators with Arduino – Part II @ 04:30.**

**QUESTION 14:**

Which device provides controlled rotary motion?

- a. Ultrasonic sensor
- b. DHT sensor
- c. PIR sensor
- d. Servo motor

**Correct Answer:** d. Servo motor

**Detailed Solution:** A servo motor is an actuator that provides precise and controlled rotary motion.

**See Lecture 25: Integration of Sensors and Actuators with Arduino – Part II @ 04:30.**



**QUESTION 15:**

State whether the following statement is True or False.

**Statement:** A stepper motor is a type of actuator.

- a. True
- b. False

**Correct Answer:** a. True

**Detailed Solution:** A stepper motor is one of the motor actuators discussed in the lecture. Other examples include servo motors, hydraulic motors, solenoids, relays, and AC motors.

**See Lecture 25: Integration of Sensors and Actuators with Arduino – Part II @ 03:20.**

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